A Deep Learning–Based Intelligent Framework for Road Safety Monitoring and Traffic Violation Detection

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***Abstract*—Globally, With the rapid increase in vehicles, traffic rule violations have become a major cause of accidents, congestion, and inefficiency in urban environments. Traditional traffic monitoring systems rely heavily on manual surveillance, which is time-consuming, error-prone, and difficult to scale. Recent advancements in computer vision and deep learning have enabled automated and real-time detection of traffic violations using video surveillance systems. This survey report reviews existing research and technologies related to automated traffic violation detection, with a focus on helmet detection, no- parking detection, lane violation detection, and vehicle tracking systems. The report also presents the current progress of our proposed system, which integrates d–eep learning-based object detection with multi-zone no-parking violation detection and stationary vehicle analysis. The study highlights the strengths, limitations, and research gaps in existing systems and motivates the need for a robust, scalable, and real-time intelligent traffic monitoring framework.**

Keywords*— *Helmet detection, Number plate detection, No parking Detection YOLOv8, CNN, Road Safety, Automated System, Vehicle Tracking System, Intelligent Traffic Monitoring

I. Introduction

Traffic congestion and road safety have become major concerns in rapidly growing urban environments due to the increasing number of vehicles. Among various modes of transportation, two-wheelers are widely preferred because of their affordability and ease of use. However, the growing number of motorcycles on the roads has also led to a significant increase in accidents. A major factor contributing to these accidents is irresponsible riding and the failure to follow safety protocols, particularly the neglect of helmet usage. The absence of helmets often results in severe head injuries, which are among the leading causes of fatalities in road accidents. This highlights the urgent need for strict enforcement of traffic safety regulations.

In addition to helmet violations, other traffic offenses such as triple-seat riding, signal jumping, unauthorized vehicle entry in restricted zones, and illegal parking contribute to traffic mismanagement and road hazards. Traditional traffic monitoring systems rely heavily on manual supervision by authorities, which is time-consuming, labor-intensive, and prone to human error. Such systems are inefficient in handling multiple violations simultaneously, especially in densely populated urban areas. Although certain solutions such as sensor-based detection systems have been proposed, their high cost and limited scalability make them impractical for widespread implementation.

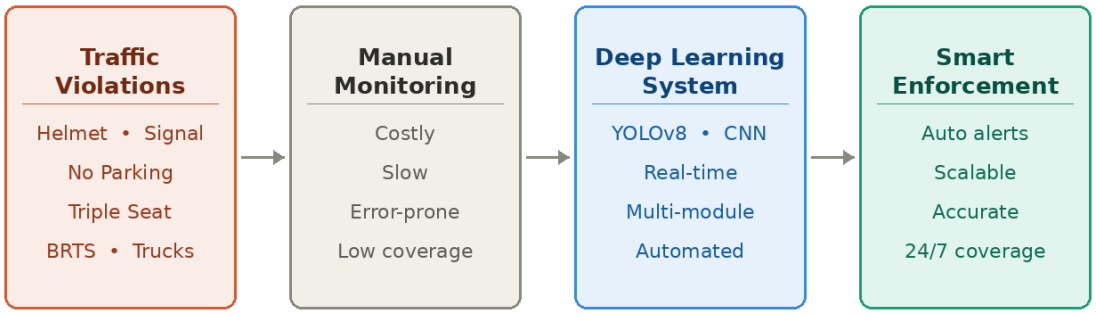

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Fig.1. Transition from Manual Traffic Monitoring to Deep Learning-Based Violation Detection System

With advancements in deep learning and computer vision, automated traffic monitoring has become a feasible and efficient solution. Modern object detection techniques based on convolutional neural networks (CNNs) enable real-time detection and classification of vehicles and traffic violations. In this work, a YOLOv8-based multi-violation detection system is proposed to identify various traffic violations, including helmet non-compliance, triple-seat riding, truck entry in restricted time zones

The proposed system integrates all detection modules into a unified framework capable of processing live video feeds and detecting multiple violations simultaneously. By automating the detection process, the system reduces dependence on manual monitoring while improving accuracy and efficiency. This approach provides a scalable and cost-effective solution for intelligent traffic management and supports the development of smart city infrastructure.

II. Literature Review

Recent advancements in deep learning and computer vision have significantly contributed to the development of automated traffic monitoring systems. These technologies enable real-time detection of vehicles and violations, improving road safety and reducing manual intervention. Traditional traffic monitoring methods rely heavily on human supervision, which is time-consuming, error-prone, and inefficient in handling large-scale traffic scenarios. As a result, there is a growing need for intelligent systems capable of detecting multiple traffic violations simultaneously.

Several research studies have explored the use of YOLO-based object detection models for traffic violation detection. Shankar K. et al. proposed a system using YOLOv9 integrated with convolutional neural networks and centroid tracking to detect helmet violations and recognize number plates [1]. The system achieved high accuracy under normal conditions but faced challenges in low-light environments and with unclear images. Similarly, Comus Bala et al. developed a real-time wrong-side driving detection system using YOLOv8 combined with SORT tracking and OpenCV techniques [2]. Although effective, the system required high computational resources and showed limitations in complex traffic conditions.

Further research by Anas Bin Malik et al. introduced a multi-stage helmet detection system using YOLOv8 and YOLOv11 models [3]. This approach improved detection accuracy through layered processing; however, issues such as occlusion and dataset imbalance affected performance. In another study, K. Kumaran et al. proposed a hybrid approach combining YOLO and Haar Cascade for license plate recognition [4]. While the system achieved high accuracy, it faced challenges under varying environmental conditions and increased memory usage. Additionally, Senthil Kumar A. et al. developed a YOLOv10-based framework for license plate recognition of non-helmeted riders, achieving promising results but depending heavily on dataset quality and GPU resources [5].

From the reviewed literature, it is evident that YOLO-based models play a crucial role in modern traffic monitoring systems due to their real-time detection capabilities and high accuracy. However, most existing systems focus on detecting individual violations and lack integration of multiple modules within a single framework. Challenges such as false positives, environmental sensitivity, and computational complexity remain unresolved.

To address these limitations, the proposed system introduces a YOLOv12-based multi-violation detection framework capable of detecting helmet violations, triple-seat riding, no-parking violations, truck restrictions, signal jumping, and BRTS lane violations in real time. The integration of multiple detection modules into a unified system enhances efficiency, scalability, and overall performance, making it suitable for intelligent traffic management applications.

III. Proposed System

A. Overview

The proposed system is a YOLOv8-based multi-violation traffic detection framework designed to provide an automated and intelligent solution for monitoring road safety violations in real time. The system processes video input from CCTV cameras or recorded traffic footage and analyzes each frame using deep learning techniques to detect vehicles, riders, and other traffic elements. By leveraging the capabilities of the YOLOv8 model, it ensures fast and accurate object detection, enabling efficient real-time analysis of complex traffic scenarios.

The framework integrates multiple violation detection modules into a unified pipeline, allowing simultaneous identification of various traffic violations such as helmet non-compliance, triple-seat riding, truck movement during restricted time zones,. Each module operates based on predefined rules applied to detected objects, ensuring accurate classification of violations under different environmental and traffic conditions.

The system is designed to minimize dependence on manual monitoring by automating the entire detection process and providing visual outputs with bounding boxes and violation labels for easy interpretation. It supports continuous monitoring and can be extended to generate automated reports and alerts for enforcement authorities. By combining deep learning with rule-based analysis, the system offers a scalable, efficient, and cost-effective solution suitable for smart city environments and large-scale traffic surveillance systems.

B. Individual Training Modules

1) Helmet Detection Module:
In this module, the system detects two-wheeler riders and identifies whether they are wearing helmets. The model analyzes head regions of riders and classifies them as helmet or no-helmet. If a rider is detected without a helmet, the system flags it as a violation. This module helps enforce safety regulations and reduce accident severity.

2) Triple Seat Detection Module:
This module detects the number of riders on a two-wheeler. The system counts the number of persons associated with a motorcycle and identifies cases where more than two riders are present. Such instances are flagged as triple-seat violations, helping ensure compliance with traffic safety norms.

3) Truck Restriction Detection Module:
This module detects heavy vehicles such as trucks and verifies whether they are operating within permitted time zones. If a truck is detected during restricted hours, the system marks it as a violation. This helps regulate traffic movement and reduces congestion during peak hours.

4) No Parking Detection Module:
In this module, predefined regions are monitored to detect vehicles that remain stationary for longer than a specified duration. The system identifies parked vehicles in restricted zones and flags them as violations. This helps in maintaining smooth traffic flow and avoiding congestion.

5) Signal Jumping Detection Module:
This module monitors traffic signals and vehicle movement simultaneously. The system detects vehicles crossing the stop line during a red signal and identifies them as signal-jumping violations. This enhances road discipline and reduces the risk of accidents at intersections.

C. System Interaction

The system enables automated real-time processing of video streams, allowing continuous monitoring of traffic scenarios without manual intervention. Users can load video input or connect live camera feeds, and the system processes frames dynamically using the YOLOv8 model.

The output is displayed with bounding boxes, labels, and violation tags, providing a clear visualization of detected violations. The system can also generate logs or reports containing violation details such as type, time, and location. This interaction ensures efficient monitoring and supports decision-making for traffic authorities.

D. Learning Objectives

The proposed system aims to enhance road safety and traffic management by developing an intelligent, automated framework capable of detecting multiple traffic violations in real time. It focuses on reducing dependence on manual monitoring while improving the accuracy, consistency, and efficiency of traffic rule enforcement. By utilizing deep learning techniques, the system ensures reliable detection of violations across diverse traffic conditions.

The system is designed to identify and monitor various traffic violations, including helmet non-compliance, triple-seat riding, truck entry during restricted time periods, By integrating these modules into a unified system, it enables simultaneous detection of multiple violations, thereby increasing overall monitoring efficiency and reducing operational complexity.

Additionally, the system enhances situational awareness by providing clear visual outputs such as bounding boxes, labels, and violation indicators. This helps traffic authorities quickly interpret results and take appropriate action. The system also supports continuous monitoring of traffic environments and can be extended to generate automated reports, alerts, and violation logs for further analysis.

Furthermore, the proposed framework is designed to be scalable and adaptable, allowing deployment across different traffic scenarios and infrastructure setups. By combining deep learning with rule-based decision logic, the system aims to provide a cost-effective, robust, and efficient solution for modern intelligent traffic systems, contributing to safer roads and smarter urban mobility.

IV. METHODOLOGY / IMPLEMENTATION

A. System Architecture

The proposed traffic violation detection system is implemented using a deep learning–based architecture centered around the **YOLOv8 object detection model**. The system follows a modular pipeline that processes video input from CCTV cameras or recorded footage and performs real-time detection of multiple traffic violations. The workflow begins with capturing video frames, which are then passed through a preprocessing stage to ensure consistency and improve detection accuracy.

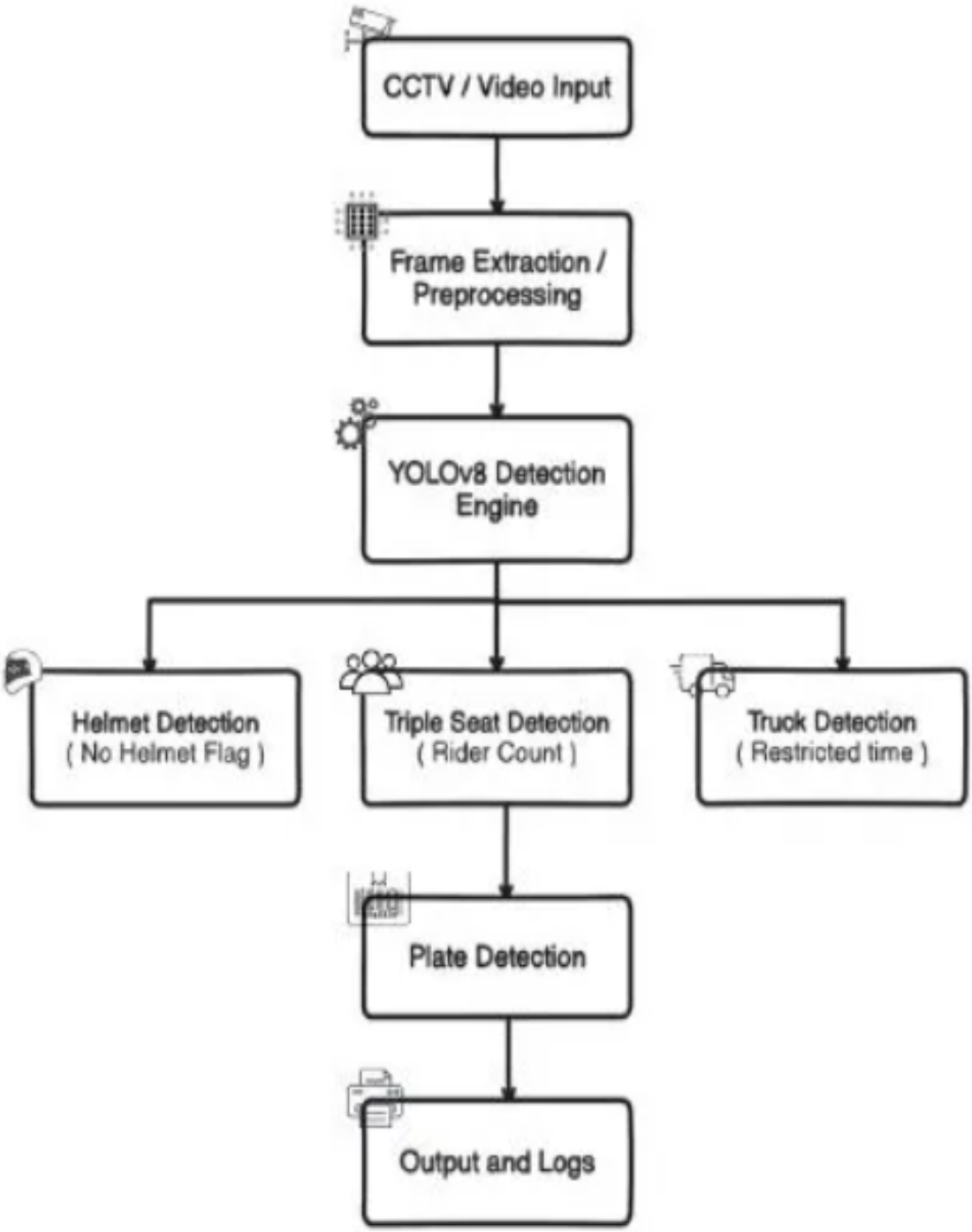

Fig.2. System Architecture

During preprocessing, each frame undergoes operations such as resizing, normalization, and image enhancement to prepare it for input into the YOLOv8 model. The processed frames are then fed into the detection layer, where the model identifies vehicles, riders, and relevant traffic elements using trained weights. The detection stage produces bounding boxes along with class labels and confidence scores for each identified object.

The architecture is designed to support multiple violation detection modules that operate simultaneously on the detected objects. Each module applies specific rule-based logic to determine violations such as helmet absence, triple-seat riding, truck restriction violations .These modules are integrated within a unified framework, allowing efficient and parallel processing of multiple traffic scenarios.

The system also includes an output layer responsible for visualizing detection results. Detected violations are highlighted using bounding boxes and labeled accordingly on the video frames. Additionally, the system can log violation details such as type, timestamp, and location for further analysis or reporting. This layered architecture ensures scalability, real-time performance, and adaptability for deployment in intelligent traffic monitoring systems.

B. Tools and Technologies

- Operating System: Windows / Linux
- Programming Language: Python 3.8+
- Libraries & Frameworks: OpenCV 4.5+, NumPy, PyTorch
- Detection Models : YOLOv8
- IDE / Development Tools: Google Colab , VS Code

These tools are used to develop a real-time traffic violation detection system. Python and its libraries enable efficient data processing and model implementation, while YOLOv8 provides high-speed and accurate object detection. OpenCV supports image and video processing, and the development environments ensure smooth implementation and testing of the system.

C. Environment Design

The system environment represents a real-time traffic monitoring setup and is divided into multiple functional components to simulate different traffic scenarios and violation detection conditions:

- **Road Monitoring Zone –** A simulated or real road segment where vehicles are detected using video input. The system observes traffic flow, identifies vehicles, and processes real-time frames for violation detection.
- **Signal Monitoring Area –** A traffic signal zone where vehicle movement is analyzed with respect to signal status. The system detects vehicles crossing stop lines during red signals to identify signal jumping violations.
- **Parking Control Zone –** A predefined area where stationary vehicles are monitored. The system checks if vehicles remain parked beyond a certain duration in restricted zones and flags no-parking violations.

The environment is designed to support real-time processing and detection under varying traffic conditions. It allows continuous monitoring, adaptability to different road layouts, and scalability for deployment across multiple locations.

D. Interaction Design

The system interacts with users and traffic data through the following mechanisms:

- **Video Input Processing –** Users can provide live CCTV feeds or recorded video input. The system processes frames dynamically and performs real-time detection of vehicles and violations.
- **Detection and Visualization –** The system highlights detected objects using bounding boxes and labels. Violations such as no helmet, triple seat, or signal jumping are clearly marked for easy interpretation.
- **System Output and Alerts –** The system generates outputs such as violation logs, timestamps, and visual alerts. These outputs can be used by traffic authorities for monitoring and enforcement purposes.
- **User Monitoring Interface –** A simple graphical interface allows users to view processed video, adjust parameters, and monitor detection results in real time.

These interactions are designed to provide an efficient and user-friendly experience, enabling accurate monitoring of traffic violations while ensuring ease of use and real-time responsiveness.

E. System Workflow

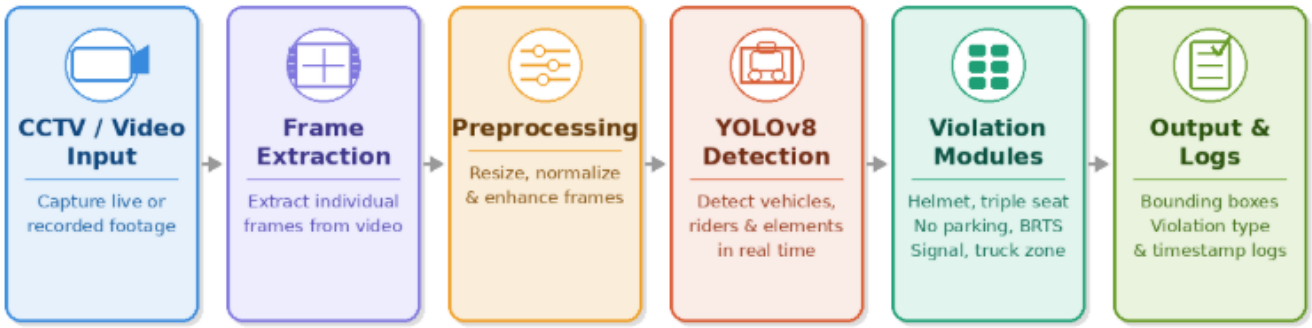

Fig. 3. Workflow of the Traffic System

The workflow of the proposed system begins with capturing video input from CCTV cameras or recorded traffic footage, which is then converted into a sequence of frames for further processing. Each frame undergoes preprocessing operations such as resizing, normalization, and noise reduction to ensure uniformity and improve the accuracy of detection. These steps help in handling variations in lighting conditions, resolution, and environmental factors.

The preprocessed frames are then passed into the YOLOv8-based detection model, which performs real-time object detection by identifying vehicles, riders, and relevant traffic components such as lanes and signal regions. The model generates bounding boxes along with class labels for each detected object. Based on these detections, the system activates multiple violation detection modules that apply rule-based logic to identify specific traffic violations.

The violation detection stage includes modules for helmet detection, triple-seat detection, truck restriction monitoring. Each module analyzes the detected objects and evaluates them against predefined rules to determine whether a violation has occurred. If a violation is detected, it is immediately flagged and highlighted on the output frame.

Following detection, the system processes the results by generating visual outputs with bounding boxes and labels for easy interpretation. Additionally, the system records relevant information such as violation type, timestamp, and frame details for further analysis or reporting. This structured workflow ensures continuous monitoring, efficient processing, and accurate detection of multiple traffic violations in real time, making the system suitable for intelligent traffic management applications.

V. RESULTS / OUTPUT

The proposed YOLOv8-based traffic violation detection system was successfully implemented and evaluated using real-time video input and recorded traffic footage. The system demonstrates the ability to detect multiple traffic violations simultaneously, including helmet non-compliance, triple-seat riding, truck movement in restricted time zonesThe results confirm that the system can process video frames efficiently and provide real-time detection outputs with reasonable accuracy.

The system generates visual outputs by marking detected objects with bounding boxes and labels, making it easier to identify violations. The integration of multiple detection modules within a single framework allows continuous monitoring of traffic scenarios without manual intervention. However, some false positives were observed under challenging conditions such as occlusion, crowded traffic, and varying lighting environments.

Module-Wise Results

1) Helmet Detection Module

The helmet detection module identifies whether a two-wheeler rider is wearing a helmet. The system successfully detects riders and classifies helmet presence or absence using the trained YOLOv12 model. In cases where no helmet is detected, the system flags it as a violation and highlights the rider with a bounding box.

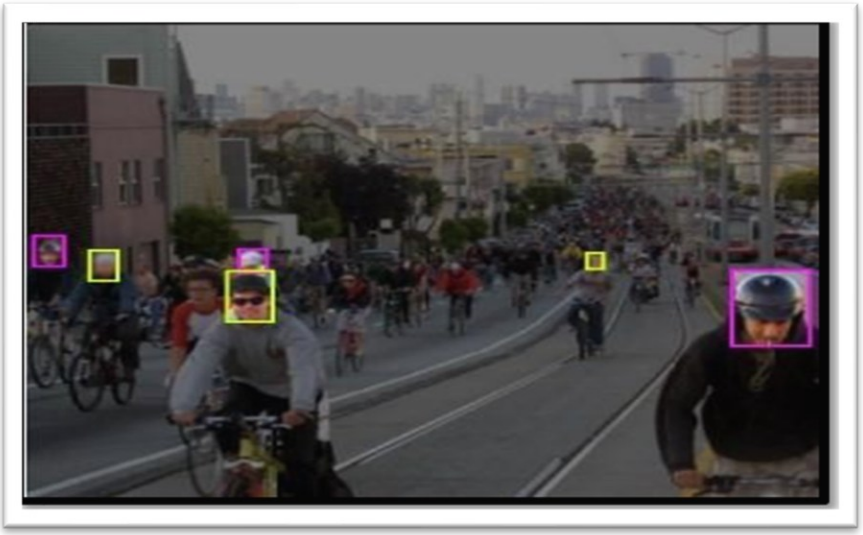

Fig 5. Helmet Detection

2) Triple Seat Detection Module

This module detects the number of riders on a motorcycle and identifies violations when more than two riders are present. The system effectively counts detected persons and flags triple-seat riding cases, helping enforce safety regulations.

Fig 6. Triple Seat Riding Detection

3) No Parking Detection Module

The no-parking module monitors predefined regions and identifies vehicles that remain stationary beyond a specified duration The system successfully detects parked vehicles in restricted zones and flags them as violations.

Fig 7. No Parking Detection

4) Truck Restriction Detection Module

This module detects heavy vehicles such as trucks and verifies whether they are operating within permitted time zones. If a truck is detected during restricted hours, the system marks it as a violation, helping regulate traffic flow.

Fig 8. Truck in Restricted Area Detection

5) Signal Jumping Detection Module

The signal jumping module monitors vehicle movement relative to traffic signals. It detects vehicles crossing the stop line during a red signal and flags such instances as violations, improving road discipline at intersections.

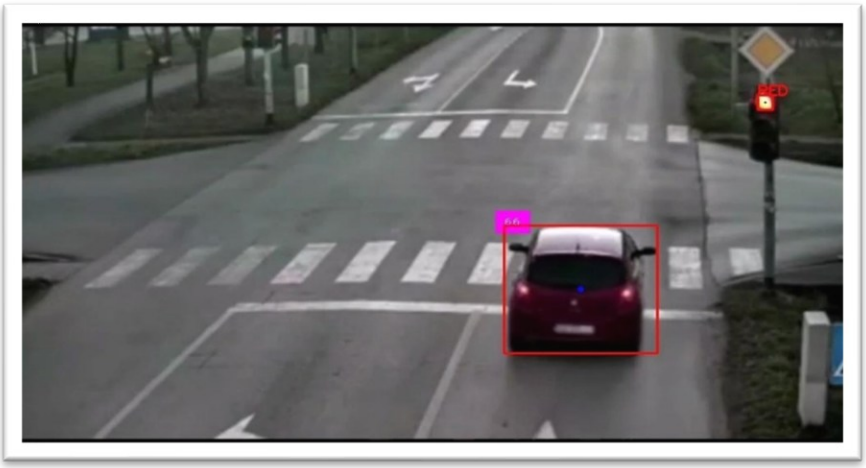

Fig 9. Signal Jumping Detection

VI. Future Work

The current implementation of the proposed YOLOv8-based traffic violation detection system focuses on real-time identification of multiple traffic violations using deep learning techniques. Although the system demonstrates effective performance, future enhancements can be made to further improve accuracy, scalability, and practical deployment.

A major extension of the system includes the integration of advanced license plate recognition (OCR) to automatically identify vehicles involved in violations. This would enable automated fine generation and improve enforcement efficiency. Additionally, incorporating multi-object tracking algorithms can enhance detection stability and reduce false positives in dense traffic conditions.

The system can also be extended to support edge computing and IoT-based deployment, allowing real-time processing directly on surveillance devices with reduced latency. Furthermore, integrating cloud-based storage and analytics can help in maintaining large-scale violation records and generating detailed reports for traffic authorities.

Future improvements may also include the use of advanced deep learning models and optimization techniques to further enhance detection accuracy under challenging conditions such as low lighting, occlusion, and adverse weather. These enhancements will contribute to making the system more robust, scalable, and suitable for smart city applications.

VII. Conclusion

This paper presents a YOLOv8-based multi-violation traffic detection system designed to automate the monitoring of road safety rules in real time. The system integrates multiple detection modules, including helmet detection, triple-seat detection, truck restriction monitoring , within a unified framework.

The proposed system successfully processes video input, detects vehicles, and identifies violations using deep learning and rule-based analysis. It provides real-time outputs with bounding boxes and violation labels, improving the efficiency and reliability of traffic monitoring compared to traditional manual methods.

Overall, the system offers a scalable, cost-effective, and intelligent solution for modern traffic management. It reduces human effort, enhances accuracy, and supports the development of smart city infrastructure. With further improvements and integration of advanced features, the system has strong potential for real-world deployment in traffic surveillance and enforcement applications.

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